

UX Research Summary

Project Title: MODU

1. Research Overview

1.1 Background and Problem

Furniture assembly is often treated as a simple practical task. However, it requires multiple cognitive processes to work together simultaneously, including:

- Planning
- Sequencing
- Working memory
- Spatial reasoning
- Sustained attention

For many people, these demands can make furniture assembly frustrating, overwhelming, or difficult to complete successfully. This includes first-time assemblers, users with limited spatial confidence, and individuals who simply find traditional assembly instructions difficult to follow.

The challenges can be even more significant for neurodivergent users, including people with:

- ADHD
- Autism Spectrum Disorder (ASD)
- AuDHD
- Dyslexia
- Dyspraxia

Across these groups, common barriers include executive dysfunction, information processing difficulties, spatial reasoning challenges, and cognitive overload.

1.2 Research Goal

- Understand barriers during furniture assembly
- Identify cognitive accessibility challenges
- Explore how adaptive learning experiences could improve task completion and confidence

1.3 Research Questions

The research will be focusing on:

- 1) What cognitive challenges do users encounter during furniture assembly?
- 2) How do individual learning preferences influence assembly behaviour and support needs?
- 3) How can personalised guidance support users during furniture assembly?

2. Key Research

Cognitive Accessibility Research (based on [cognitive dis. research](#))

Cognitive Profile	Pain Points	Potential Challenges
ADHD (Attention Deficit Hyperactivity Disorder)	• Easily distracted during assembly tasks • Difficulty maintaining attention across long instructions • Frequently loses track of progress after interruptions • May skip steps or rush through instructions • Feels overwhelmed when presented with too many parts or tasks at once	• Executive dysfunction • Working memory limitations • Difficulty initiating and completing tasks • Reduced ability to sustain attention • Increased cognitive overload under complex conditions
ASD (Autism Spectrum Disorder)	• Becomes frustrated by vague or ambiguous instructions • May struggle when multiple interpretations are possible • Experiences stress when assembly outcomes are unpredictable • Finds sudden changes or inconsistencies difficult to manage • Requires clear structure and logical sequencing	• High need for predictability and routine • Lower tolerance for uncertainty • Difficulty interpreting ambiguous information • Increased anxiety when instructions lack clarity • Preference for structured and consistent processes

AuDHD (Autism + ADHD)	• Simultaneously needs structure and flexibility • Easily loses focus while depending on clear guidance • Can become overwhelmed by either too much or too little support • Experiences fluctuating attention and cognitive load • May struggle to regulate pace throughout a task	• Combination of executive dysfunction and need for predictability • Difficulty balancing autonomy with support • Variable attention and energy levels • Increased cognitive fatigue • Complex support needs that differ across situations
Dyslexia	• Finds text-heavy instructions difficult to follow • May confuse labels, numbers, symbols, or part identifiers • Requires additional time to process written information • Needs to repeatedly check instructions for confirmation • Can become frustrated by dense instructional content	• Slower text processing • Difficulty decoding written information • Increased cognitive effort when reading • Reduced efficiency in locating key information • Higher risk of misunderstanding written instructions
Dyspraxia (Developmental Coordination Disorder)	• Struggles to understand spatial relationships between parts • Finds orientation and direction difficult to interpret • Has difficulty visualising object rotation and positioning • May confuse left/right or front/back orientations • Finds multi-step physical assembly challenging	• Spatial reasoning difficulties • Motor planning challenges • Difficulty translating instructions into physical actions • Reduced confidence when manipulating objects • Increased cognitive effort when navigating spatial tasks

Further research

While the literature review provides a strong understanding of common cognitive challenges and accessibility needs, several questions remain that require further investigation through user research and iterative design.

1. General Users and First-Time Assemblers

Many people experience frustration during furniture assembly despite having no diagnosed cognitive condition. Further research should explore:

- Common causes of frustration and task abandonment
- Confidence levels during assembly
- Previous experiences with flat-pack furniture
- Strategies users currently adopt to overcome difficulties

2. Individual Learning Preferences

Existing literature highlights the diversity of cognitive needs but does not fully explain how different users prefer to learn during assembly tasks. Future research should investigate:

- Preferred instruction formats (text, images, animation, 3D interaction)
- Preferred levels of guidance and autonomy
- User preferences for pacing and task segmentation
- Recovery strategies after interruptions or mistakes

3. Cognitive Load During Assembly

Further research is needed to understand when users experience the highest cognitive load throughout the assembly process, including:

- Reading and interpreting instructions
- Identifying and locating components
- Spatial reasoning and object orientation
- Remembering previous steps
- Recovering from errors or interruptions

4. Motivation and Engagement

As MODU incorporates elements of gamification, additional research should explore:

- Whether gamification reduces frustration or increases engagement
- Which reward mechanisms users find meaningful
- Whether motivation differs between experienced and first-time users
- The balance between guidance, challenge, and autonomy

These findings will help inform future design decisions, onboarding strategies, and the potential development of adaptive learning experiences within the product.

Resources:

[Barkley, R. A. \(2012\). Executive Functions: What They Are, How They Work, and Why They Evolved.](#) -executive dysfunction, working memory, sustained attention
[Shaywitz, S. \(2003\). Overcoming Dyslexia.](#) -text processing, decoding, reading effort

Shared Challenges

- Executive Dysfunction: inability to plan, focus, remember instructions, and juggle multiple tasks
- Spatial & Visual Processing: Traditional assembly instructions often require users to mentally transform two-dimensional diagrams into three-dimensional actions. This process places significant demands on spatial reasoning and mental rotation abilities.
- Sensory & Cognitive Overwhelm: Complex instructions, large numbers of components, and prolonged concentration demands can lead to cognitive overload, increased frustration, and, in some cases, task abandonment.

From Cognitive Categories to Support Needs

Throughout the research process, an important observation emerged. While the literature review draws on characteristics associated with ADHD, ASD, AuDHD, Dyslexia, and Dyspraxia, the recurring challenges themselves are not exclusive to any single diagnosis.

Instead, many users, including first-time assemblers and those without a diagnosed cognitive condition, may experience similar difficulties with executive function, information processing, spatial reasoning, or the need for clear and predictable guidance.

For this reason, the project should avoid framing its design around medical categories. Instead, these cognitive characteristics should serve as research inspiration for identifying broader support needs and interaction patterns.

From a UX perspective, designing for support needs rather than diagnoses promotes a more inclusive experience, avoids unnecessary labelling, and allows the system to adapt to a wider range of users. Future personalisation and avatar concepts should therefore represent different support styles or learning preferences rather than specific cognitive conditions.

Based on the literature review, four recurring themes emerged across multiple cognitive profiles. These themes provide a more practical framework for understanding user needs and may serve as a foundation for future learning profile development:

1. Executive Function Challenges
2. Information Processing Challenges
3. Spatial Reasoning Challenges
4. Predictability & Cognitive Safety

Theme	Related Profiles	Common Challenges
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Executive Function Challenges	ADHD, AuDHD etc.	• Maintaining attention • Task initiation • Progress tracking • Managing interruptions
Information Processing Challenges	Dyslexia, ADHD etc.	• Processing large amounts of information • Identifying key instructions • Managing cognitive load
Spatial Reasoning Challenges	Dyspraxia, ASD (some users) etc.	• Understanding object orientation • Mental rotation • Translating 2D instructions into 3D actions
Predictability & Cognitive Safety	ASD, AuDHD etc.	• Need for structure • Need for clarity • Anxiety around mistakes • Difficulty with uncertainty

Existing Solutions Review

The competitive analysis examined products across four key categories: room planning and visualisation, assembly guidance, gamified learning, and avatar-based engagement systems. (from [Competitors Analysis](#))

Category	Focus	Product	Platform	Links
Assembly Solutions	Furniture and space cognitive 3D interaction assembly workflow Assembly guidance	IKEA Kreativ	Web + IKEA app	https://www.ikea.com/ie/en/home-design/dashboard/
		Home improvisation	Desktop (PC / Steam)	http://www.homeimprovisation.com/
		Furniture assembly simulator	Desktop (PC / Steam)	https://store.steampowered.com/app/4395740/Furniture_Assembler_Simulator/
		Assemble with Care	Mobile (iOS & Android)	https://play.google.com/store/apps/details?id=com.ustwo.awc&hl=en_IE
		BILT	Mobile (iOS & Android)	https://play.google.com/store/apps/details?id=com.bilt.mobile&hl=en
Room Planning & Decoration	room ownership decoration progression emotional engagement	Catroom	Mobile (iOS & Android)	https://play.google.com/store/apps/details?id=com.crossfield.catroom&hl=en_IE&pli=1
		IKEA Planner	Web	https://www.ikea.com/ie/en/planners/

Reference for Gamification Mechanics	XP progression unlock motivation retention	My Dream Setup	Desktop (PC / Steam)	https://store.steampowered.com/app/2200780/My_Dream_Setup/
		Rooms	Web	https://rooms.xyz/welcome
		Catroom	Mobile (iOS & Android)	https://play.google.com/store/apps/details?id=com.crossfield.catroom&hl=en_IE&pli=1
		Duolingo	Mobile (iOS & Android)	https://www.duolingo.cn/

Main research was focused on Kreativ and BILT, as these tools either help users visualise furniture within a space but provide little support for the assembly process itself, or improve instruction delivery but do not adapt to different cognitive needs. Additional reference products, including Catroom, Rooms, and My Dream Setup, were analysed from a gamification and engagement perspective rather than as direct competitors. These products informed ideas around long-term motivation, room customisation, progression systems, and emotional attachment to virtual spaces.

IKEA Kreativ: A room planning and visualisation tool that allows users to scan their room and place IKEA furniture in a virtual environment before purchasing. It focuses on helping users visualise how products fit within their space.

Strengths:

- Official IKEA solution with access to accurate IKEA product models
- Strong AR and spatial scanning capabilities
- Seamlessly integrated into the IKEA ecosystem
- Supports purchasing decisions through realistic room visualisation
- User-friendly and accessible for general consumers

Weaknesses:

- Focuses only on pre-purchase planning
- Provides no support for furniture assembly after purchase
- Does not address cognitive accessibility needs
- Lacks personalised guidance or learning support
- No gamification or motivation mechanisms

BILT: An interactive 3D instruction platform that replaces traditional paper manuals with step-by-step digital assembly guides. Users can rotate, zoom, and follow animated instructions during assembly.

Stage	User Action	What BILT Provides	UX Observation	Limitation / Opportunity for MODU
App opening	Opens BILT	Intro pop-up: "Build. Keep. Track."	Quickly explains product purpose	Functional intro, but no personalisation
Product search	Searches brand / furniture	Search bar for brand or product	Efficient if user knows exact product	Discovery depends on knowing what to search
Product overview	Selects furniture model	Shows required time, people, and number of steps	Helps set expectations before starting	Time may show N/A, which weakens planning support
Model download	Downloads instruction model	Loading/download process	Makes model available for guided viewing	Waiting time creates small friction
Assembly mode	Turns phone horizontally	Landscape 3D instruction interface	Better screen space for 3D model	User must adapt to fixed orientation
Step guidance	Follows steps while assembling real furniture	Step-by-step 3D animation and component info	Clearer than paper manuals	User watches and copies; not interactive assembly
Model interaction	Zooms and rotates model	Pinch zoom and rotation	Supports spatial understanding	Cannot move or assemble parts freely
Task completion	Completes each real-world step	App tracks progress through instructions	Useful as a guided manual	No adaptive guidance, feedback, or personalised support

Strengths:

- High-quality 3D assembly instructions
- Interactive viewing (rotate, zoom, replay steps)
- Clearer than traditional paper manuals

- Widely adopted by furniture and hardware manufacturers
- Reduces ambiguity compared to static instructions

Weaknesses:

- Functions primarily as a digital manual
- Limited support for cognitive accessibility
- No adaptive learning or personalised guidance
- Does not actively reduce cognitive load
- No gamification, motivation, or engagement system
- Focuses on instruction delivery rather than skill development

Key takeaway: Existing solutions focus on visualisation, instructions, or engagement, but do not support personalised learning during assembly.

3. Research Findings

Based on the cognitive accessibility research and competitor analysis, three key findings exist as follows:

Finding 1: Furniture assembly is a cognitive challenge

Furniture assembly requires users to simultaneously maintain attention, process instructions, remember information, plan ahead, and reason spatially. For many users, particularly those experiencing executive function, information processing, or spatial reasoning challenges, these cognitive demands can create significant barriers throughout the assembly process.

Finding 2: Users require different levels and types of support

The research indicates that cognitive challenges manifest differently across individuals. Some users benefit from highly structured, step-by-step guidance, while others prefer greater autonomy and flexibility. This suggests that a single instruction format may not effectively support diverse learning needs.

Finding 3: Existing solutions focus more on completing tasks

Current solutions such as IKEA Kreativ and BILT successfully support visualisation and instruction delivery. However, they primarily focus on helping users complete tasks rather than helping users learn how to complete tasks according to their individual needs and abilities.

Key Opportunity

The gap identified through this research is not a lack of assembly instructions, but a lack of personalised learning support during assembly.

Rather than asking “What should the user do next?”, this project explores “How should this particular user learn the next step”. This shift from instruction delivery to adaptive learning forms the foundation for the next stage of the project.

Target Users

Primary

- General first furniture assembly users with inclusive design that also helps **Neurodivergent People including: ADHD, ASD, AuDHD, Dyslexia, Dyspraxia**

Secondary

- Parents & Caregivers
- Occupational Therapists
- Interior Designers
- Supported Living Staff

4. Conclusion

For many individuals, particularly those with executive function, information processing, or spatial reasoning challenges, these demands can create frustration, reduce confidence, and increase the likelihood of task abandonment.

The findings also indicate that users do not require the same type or level of support. Different individuals may benefit from different instruction formats, pacing, and degrees of guidance. This highlights a limitation of existing assembly solutions, which typically provide a fixed instruction system regardless of user needs.

Competitor analysis revealed that current products successfully address room visualisation, assembly guidance, or user engagement in isolation. However, few solutions focus on how users learn during furniture assembly or adapt support according to individual cognitive needs.

Taken together, these findings suggest that the key opportunity lies not in providing more instructions, but in providing more personalised support. Rather than treating furniture assembly as a standardised task, there is potential to approach it as a learning experience that adapts to different users, abilities, and levels of confidence.

This conclusion forms the foundation for the next stage of the project, which explores how adaptive guidance and personalised learning profiles can support a more accessible furniture assembly experience.

1. Problem



2. Cognitive Accessibility Research



3. User Findings



4. Learning Profile Framework



5. Adaptive Task Segmentation



6. Product Features



7. User Journey



8. Prototype Direction



9. Future Testing Plan